

Title	Memo text	Origin	Modified by	Modified	Created by	Created	Document name	Code	Document group	Document set	Beginning	End
Circular strategy	Major code group for the type of circular intervention discussed in the source.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Circular strategy			
Maintenance	Action that preserves the condition and performance of an existing window system without major replacement. Use when the source discusses regular inspection, cleaning, repainting, sealing, adjustment or updates. Exclude when the action replaces a major component or changes the system performance level. Example: periodic maintenance of wooden windows to prevent deterioration.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Maintenance			
Repair	Corrective action that fixes damage or failure in an existing component. Use when a damaged part is fixed so the original window or component can remain in use. Exclude routine maintenance or full replacement. Example: repairing damaged seals, hinges, glazing beads or frame joints.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Repair			
Component replacement	Replacement of a short-life or damaged part while retaining the larger window system. Use when the source discusses replacing seals, gaskets, glazing units, coatings, hardware or other parts. Exclude whole-window replacement. Example: replacing an IGU edge seal instead of discarding the entire doublet unit.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Component replacement			
Upgrade	Improvement of performance while keeping most of the existing window system. Use when the source discusses energy retrofit, secondary glazing, storm windows or improved airtightness. Exclude full removal of the existing system. Example: adding secondary glazing to improve thermal performance.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Upgrade			
Refurbishment	More extensive removal of an existing window or component to restore function, appearance or performance. Use when the source discusses restoration, remanufacturing or returning a used component to serviceable condition. Exclude minor repair only. Example: restoring historic steel windows instead of replacing them.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Refurbishment			
Direct reuse	Use of an existing component again with little or no processing. Use when the source discusses reusing a component in the same or similar function. Exclude recycling, downcycling or material recovery. Example: reusing a recovered window frame or facade component.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Direct reuse			
Recycling	Processing material into secondary raw material after the component is no longer used. Use when the source discusses material recovery, recycling rates, recycling credits or end-of-life processing. Exclude intact component reuse. Example: recycling aluminium window frame material.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Recycling			
Obsolescence	Major code group for the reason a window system or component loses usefulness, value or desirability.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Obsolescence			
Physical	Loss of function due to material ageing, damage, decay or wear. Use when deterioration, corrosion, fading, surface degradation or mechanical damage is the issue. Exclude market preference, regulation or outdated performance expectations. Example: timber decay, seal failure or corroded steel frames.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Physical			
Functional	Loss of adequacy because the window no longer meets current performance needs. Use when the issue is thermal comfort, airtightness, acoustic performance, operability or energy performance. Exclude physical damage when performance remains adequate. Example: existing windows replaced because they are considered thermally inadequate.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Functional			
Technical	Loss of usefulness because the design or technology prevents repair, upgrade or recovery. Use when components are bonded, inaccessible, inseparable or technically hard to disassemble. Exclude simple physical damage. Example: insulated glazing units cannot be refurbished because the edge seal is not recoverable.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Technical			
Economic	Loss of usefulness because repair, reuse or maintenance is not financially attractive. Use when labour cost, service cost, storage cost, lack of market value or investment cost is the barrier. Exclude technical impossibility. Example: replacement chosen because repair appears too expensive or uncertain.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Economic			
Regulatory	Loss of usefulness because the product no longer meets legal, safety, energy or certification requirements. Use when the source discusses regulation, certification, standards, warranty or compliance barriers. Exclude regulation mentioned only as general context. Example: reused components cannot be accepted because certification is missing.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Regulatory			
Aesthetic	Loss of desirability because appearance, style or architectural preference changes. Use when replacement is driven by visual preference, façade image, modernization or perceived outdated appearance. Exclude actual damage or performance failure. Example: historic windows replaced because new windows are visually preferred.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Aesthetic			
User-driven	Loss of usefulness because user behaviour, expectations or comfort preference change. Use when occupants, owners or users change demand for comfort, convenience, operation or appearance. Exclude changes caused mainly by technical failure. Example: users replace windows to achieve higher comfort even when repair is possible.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			User-driven			
Enabling condition	Major code group for conditions that must exist for a circular strategy to work in practice.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Enabling condition			
Component accessibility	Components can be reached for inspection, maintenance, repair or replacement. Use when accessibility is necessary for the strategy to work. Exclude general repair without access issues. Example: seals and hardware are accessible for replacement.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Component accessibility			
Disassembly	The system can be taken apart without destroying valuable components. Use when the source discusses reversible connections, disassembly sequence, selective deconstruction or design for disassembly. Exclude demolition without recovery. Example: façade components removed in a planned sequence.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Disassembly			
Spare parts	Replacement parts are available for maintenance, repair or component replacement. Use when the source discusses gaskets, seals, fittings, coatings or other parts needed over time. Exclude general material availability only. Example: spare seals available for future window repair.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Spare parts			
Repair information	Information exists about materials, connections, maintenance history or repair procedure. Use when drawings, manuals, material passports, BIM data or disassembly guides are needed. Exclude repair that is possible without information barriers. Example: material passport records the window composition and connection types.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Repair information			
Skilled labour	Qualified people are available to inspect, repair, refurbish or disassemble the system. Use when expertise, workmanship or specialist knowledge affects the strategy. Exclude labour mentioned only as cost. Example: preservation professionals restore historic steel windows.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Skilled labour			
Service infrastructure	Local systems exist for repair, refurbishment, reuse, reverse logistics, storage or recycling. Use when the source discusses practical infrastructure outside the product itself. Exclude product design only. Example: local repair and refurbishment services support window service life extension.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Service infrastructure			
Business model	A commercial or organisational model supports the circular strategy. Use when the source discusses service models, supplier responsibility, take-back, leasing, refurbishment services or market demand. Exclude technical feasibility only. Example: supplier offers long-term repair and component replacement.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Business model			
Regulation	Policy, rules or public requirements support the strategy. Use when the source discusses policy support, legal requirements or regulatory drivers. Exclude regulation mentioned only in passing. Example: regulation encourages reuse or repair instead of disposal.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Regulation			
Certification	Certification, testing or warranty systems make reused or repaired components acceptable. Use when the source discusses certification barriers, performance validation or acceptance of secondary components. Exclude general quality only. Example: reclaimed façade parts need certification before reuse.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Certification			
User acceptance	Users, owners or clients accept the circular strategy. Use when preferences, trust, appearance, comfort or perceived quality affect adoption. Exclude technical performance only. Example: owners accept repaired windows instead of demanding full replacement.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			User acceptance			
Negative mechanism	Major code group for mechanisms through which a circular strategy may create unintended consequences or reduce expected environmental benefits.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Negative mechanism			
Premature replacement	A window or component is replaced before the end of its useful service life. Use when repair, maintenance or lower-material intervention may have been sufficient. Exclude clearly necessary replacement due to severe failure. Example: full window replacement despite repairable defects.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Premature replacement			
Frequent upgrading	Products are upgraded more often because better performance or new technology becomes available. Use when repeated upgrades may reduce the environmental benefit. Exclude one justified retrofit. Example: replacing windows repeatedly for material efficiency improvements.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Frequent upgrading			
Increased material complexity	A strategy adds layers, materials or assemblies that make future repair, recycling or recovery harder. Use when coatings, laminated layers, adhesives, cladding or composite assemblies create complexity. Exclude simple material substitution without recovery implications. Example: added coatings improve performance but reduce recyclability.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Increased material complexity			
Additional transport	Circular strategy creates extra transport for repair, reuse, refurbishment, storage or recycling. Use when logistics may reduce expected benefit. Exclude negligible or irrelevant transport. Example: recovered components require long-distance transport to a reuse facility.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Additional transport			
Higher initial material use	A strategy requires more material at the beginning to improve durability, performance or future circularity. Use when added material may offset later savings. Exclude unchanged material use. Example: aluminium cladding reduces maintenance but increases production impact.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Higher initial material use			
Increased comfort or use	Efficiency or circular improvements lead to more use, higher comfort expectations or increased demand. Use when behavior may offset environmental savings. Exclude comfort improvement without evidence of increased use. Example: better windows allow higher indoor temperatures and reduce heating sources.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Increased comfort or use			
Burden shifting	Impact is moved to another lifecycle stage, actor, location or environmental category. Use when a benefit in one area causes harm elsewhere. Exclude simple uncertainty. Example: lower carbon impact comes with higher toxicity impact from coating choice.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Burden shifting			
Lock-in	A long-lived solution restricts future repair, upgrading, disassembly or environmental improvement. Use when current design choices make later circularity harder. Exclude temporary barriers that can easily be changed. Example: permanent seals prevent future IGU refurbishment.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Lock-in			
Missing repair infrastructure	Repair, refurbishment or reuse fails because the needed services, skills or logistics are unavailable. Use when the strategy depends on a practical system that does not exist. Exclude barriers caused purely by product design. Example: repair is possible in theory but no local service provider exists.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Missing repair infrastructure			
Mitigation	Major code group for interventions that reduce rebound risk, burden shifting, implementation failure, lifecycle trade-offs or technological lock-in.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Mitigation			
Design intervention	A design change reduces the risk of unintended consequences. Use when the source suggests redesign, separable parts, reversible connections or accessible components. Exclude policy or business measures. Example: design the IGU edge seal as replaceable.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Design intervention			
Maintenance planning	Planned inspection and maintenance reduces premature failure or replacement. Use when schedules, inspections, defect classification or maintenance strategies are proposed. Exclude one-time repair only. Example: regular defect inspections for wooden windows.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Maintenance planning			
Supplier commitment	Suppliers or manufacturers support long-term repair, parts, take-back or product stewardship. Use when long-term responsibility is needed from the product provider. Exclude general manufacturer involvement. Example: manufacturer provides spare assets and repair guidance.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			Supplier commitment			
User guidance	Users or owners receive instructions that support correct use, maintenance or repair decisions. Use when behaviour, acceptance or correct use matters. Exclude technical documentation aimed only at professionals. Example: owners receive guidance on when to repair instead of replace.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49			User guidance			

Business-model intervention	A business model supports circular action and reduces practical barriers. Use when service contracts, leasing, repair services or take-back systems are proposed. Exclude product redesign alone. Example: maintenance service model supports window life extension.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49	Business-model intervention
Policy intervention	Regulation, incentives or public requirements reduce barriers or improve adoption. Use when policy support is needed to make the circular strategy work. Exclude voluntary business action only. Example: policy supports repair, reuse certification or landfill diversion.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49	Policy intervention
Lifecycle verification	Environmental claims are checked through LCA, LCC, sensitivity analysis or scenario testing. Use when the source recommends verifying whether the expected benefit is real. Exclude circular benefit assumed without evidence. Example: compare maintenance and replacement scenarios before decision.	Code	user	22.6.2026 r. 12:46:49	user	22.6.2026 r. 12:46:49	Lifecycle verification